

SYNTHETIC DATA PREPARED FOR AGRONET

Generic Pest & Disease Management Handbook for Indian Agriculture

This handbook is intended for educational purposes and RAG testing. It provides general guidance on integrated pest and disease management practices commonly recommended in Indian agriculture. Management decisions should always be based on local agricultural advisories, field observations, and product labels.

1. Introduction

Crop losses due to insects, diseases, weeds, rodents, and other organisms can reduce yield and quality. An integrated approach combining prevention, monitoring, biological methods, cultural practices, and responsible pesticide use provides the most sustainable management strategy.

2. Types of Crop Pests

Major pest groups include:

- Chewing insects
- Sap-sucking insects
- Borers
- Leaf miners
- Mites
- Nematodes
- Rodents
- Birds in selected cropping systems

3. Types of Plant Diseases

Common disease categories:

- Fungal diseases
- Bacterial diseases
- Viral diseases
- Phytoplasma diseases
- Nematode-associated diseases

Disease development often depends on a susceptible host, a disease-causing organism, and favorable environmental conditions.

4. Integrated Pest Management (IPM)

Core IPM principles:

- Regular field scouting
- Early detection
- Economic threshold-based decisions

- Biological control
- Mechanical and physical control
- Cultural management
- Responsible pesticide application only when necessary

5. Field Monitoring

Inspect fields regularly for:

- Leaf damage
- Stem damage
- Wilting
- Yellowing
- Spots or lesions
- Insect eggs and larvae
- Beneficial insects

Maintain records of observations and weather conditions.

6. Cultural Management Practices

Preventive measures include:

- Crop rotation
- Field sanitation
- Timely sowing
- Balanced fertilization
- Proper irrigation
- Removal of infected plant debris
- Weed management
- Use of healthy planting material

7. Biological Control

Useful biological agents include:

- Ladybird beetles
- Lacewings
- Predatory spiders
- Parasitoid wasps
- Trichoderma species
- Bacillus-based microbial products

Conserving beneficial organisms reduces unnecessary pesticide use.

8. Mechanical & Physical Control

Examples:

- Hand removal of infected plants
- Light traps

- Sticky traps
- Pheromone traps
- Bird perches
- Netting and barriers
- Deep summer ploughing where appropriate

9. Responsible Pesticide Use

Follow these practices:

- Read the product label carefully.
- Apply only recommended doses.
- Wear gloves, masks and protective clothing.
- Avoid spraying during strong winds.
- Prevent contamination of water bodies.
- Observe the pre-harvest interval.
- Rotate pesticide groups to reduce resistance.

10. Disease Prevention

General recommendations:

- Use certified seed.
- Treat seed where recommended.
- Improve drainage.
- Avoid excessive irrigation.
- Maintain balanced nutrition.
- Remove severely infected plants.
- Clean farm equipment after use.

11. Weather and Pest Risk

High humidity may favor fungal diseases.

Extended dry periods may increase some insect infestations.

Heavy rainfall can spread certain diseases.

Regular monitoring after major weather events is recommended.

12. Safe Storage and Disposal

- Store pesticides in original containers.
- Keep away from food and animal feed.
- Lock storage areas.
- Dispose of empty containers according to local regulations.
- Never reuse pesticide containers for household purposes.

13. Best Practices Checklist

Before Season:

- Clean field

- Select healthy seed
- Plan crop rotation

During Season:

- Scout weekly
- Identify pests correctly
- Encourage beneficial insects
- Record observations

After Harvest:

- Remove crop residues where appropriate
- Clean equipment
- Review pest records for future planning

14. Summary

Integrated Pest Management combines prevention, observation, biological methods, cultural practices, and careful pesticide use. Regular monitoring and timely action help reduce crop losses while protecting beneficial organisms, the environment, and farm sustainability.